

THOMAS MARKHORST

Amsterdam, The Netherlands

Phone: +31 (6) 11052279 ◊ Email: thomas.c.markhorst@gmail.com ◊ 

PhD candidate in Computer Vision at TU Delft developing multimodal and generative models for human perception and understanding. My research spans 2D/3D human pose estimation, video and motion understanding, diffusion models, and multimodal large language models. I am particularly interested in robust human-centric perception and reasoning from multimodal data, including human pose, motion, behavior, and interactions in real-world environments.

EDUCATION

PhD Computer Vision - TU Delft (NL)

Mar. 2024 – Mar. 2028

- Researching multimodal and generative models for human perception and understanding
- Developed diffusion models with temporal and inter-person modeling for 3D pose estimation and integrated pose representations into a pretrained Video-LLM using staged fine-tuning for fine-grained motion reasoning
- Head teaching assistant for the Computer Vision MSc course, coordinating 8 teaching assistants and 100+ students; supervised 2 MSc and 5 BSc theses

Master Artificial Intelligence - TU Delft (NL)

Sept. 2021 – Aug. 2023

- Cum laude & thesis published at ECCVw '24

Thesis: 9.0; GPA: 8.7

Bachelor Computer Science - TU Delft (NL)

Sept. 2018 – July 2021

- Cum laude & honours program

Thesis: 9.5; GPA: 8.9

PROFESSIONAL EXPERIENCE

Bosch Computer Vision Research Intern

(Eindhoven, NL) Dec. 2022 – Aug. 2023

- Developed joint image denoising and classification models for robust recognition in challenging light
- Designed latency-aware neural architecture search methods to optimize models for edge devices

BMW Group Computer Vision Research Intern

(Munich, DE) Aug. 2022 – Nov. 2022

- Adapted SOTA pose estimation algorithms to stereo vision, reducing reliance on dedicated depth sensors
- Developed a 3D synthetic data generation pipeline, reducing the need for manually labeled data

Krill Robotics Computer Vision Engineer

(Delft, NL) Feb. 2019 – Sept. 2021

- Developed the MVP vision system for object avoidance and helped secure government funding

Dutch Org. for Applied Science Software Engineering Intern

(The Hague, NL) Feb. 2017 – Mar. 2018

- Developed a VR-controlled robotic vehicle for remote presence and operation in hazardous environments

PUBLICATIONS

† FIRST AUTHOR ‡ SECOND AUTHOR

Fine-grained Human Motion Understanding with Language Models †

Under review

- Integrates 2D/3D pose representations with a Video-LLM for fine-grained human motion understanding

MuPPet: Multi-Person 2D-to-3D Pose Lifting using Diffusion †

CVPRw '26

- Uses diffusion with temporal and inter-person modeling for multi-person 3D pose estimation

PolySLGen: Online Multimodal Speaking–Listening Reaction Generation ‡

CVPR '26

- Generates speech and body motion in group interactions using pose fusion and social cue modeling

Pushing Joint Image Denoising and Classification to the Edge †

ECCVw '24

- Uses latency-aware neural architecture search for joint image denoising and classification

ADDITIONAL INFORMATION

Technical Languages	Python, PyTorch, NumPy, Git, OpenCV, Docker, ROS, C++, Blender
Interests	English (Fluent), Dutch (Native), German (Intermediate), Greek (Beginner) Sailing, Running, Hiking, Skiing, Kitesurfing, Glassworking, Goldsmithing